M11 Review / Repair - Teaching RC2

OPEN ONLY AFTER SAVED ATTEMPT

P01 repair - Row vs columnar storage; where Parquet fits

Storage layout changes how much data an analytical query must read.

Repair principle: CSV is text/row-like and convenient for exchange/debugging but weakly typed.

Key proof: database row store -> transactional point reads/writes

STOP - check your understanding

Close Review and solve a changed case.

Answer in your own words before continuing.

P02 repair - Parquet schema, row groups and metadata

Parquet performance/correctness lives partly in metadata. You should be able to inspect what was written without guessing.

Repair principle: Schema stores field names/types.

Key proof: metadata -> compression/statistics inspectable

STOP - check your understanding

Close Review and solve a changed case.

Answer in your own words before continuing.

P03 repair - Projection, predicates and analytical reads

Reading all columns/rows wastes I/O and memory when a query needs a small subset.

Repair principle: Projection means selecting only required columns.

Key proof: row-group skipping -> metadata-based chunk skip

STOP - check your understanding

Close Review and solve a changed case.

Answer in your own words before continuing.

P04 repair - Compression, small files and compaction

Thousands of tiny files create metadata/list/open overhead and poor scan efficiency even when total bytes are modest.

Repair principle: Parquet applies encodings/compression per column/page; similar column values often compress well.

Key proof: storage layout -> new files

STOP - check your understanding

Close Review and solve a changed case.

Answer in your own words before continuing.

P05 repair - Partitioning strategy and over-partitioning

Partition directories can skip large groups quickly, but high-cardinality partitions create tiny files/directories and metadata overhead.

Repair principle: Partitioning physically groups files by values such as date/region.

Key proof: status -> can be skewed/low benefit

STOP - check your understanding

Close Review and solve a changed case.

Answer in your own words before continuing.

P06 repair - Lake vs warehouse vs lakehouse

Architecture choices affect cost, open formats, transactions, BI performance and governance.

Repair principle: A data lake stores files/object data flexibly, often open formats and multiple data types.

Key proof: lakehouse -> open-ish lake files + table transactions/metadata

STOP - check your understanding

Close Review and solve a changed case.

Answer in your own words before continuing.

P07 repair - Delta tables, transaction log and ACID concepts

Plain file rewrites can leave readers seeing partial or inconsistent sets. A transaction log records which files form each committed table version.

Repair principle: Delta table commonly stores Parquet data files plus **_delta_log** transaction log.

Key proof: manual file deletion -> unsafe

STOP - check your understanding

Close Review and solve a changed case.

Answer in your own words before continuing.

P08 repair - MERGE/upsert and current-state records

Incremental sources contain inserts and updates. Blind append creates multiple current rows for one key.

Repair principle: MERGE matches source rows to target by a business key.

Key proof: older update -> does not overwrite if policy enforces version

STOP - check your understanding

Close Review and solve a changed case.

Answer in your own words before continuing.